Automated Identification of Printed Circuit Board Mount Layouts Under Process-Strain and Lead Dynamic Stress Constraints

Abstract— PCB mounting locations, often dictated by packaging constraints, significantly influence local board deformation under process loading and the transfer of motion to heavy component leads. This work formulates permanent mount placement for a circulator pump power electronics PCB as a staged finite element decision process for two to six active mounts. A stratified maximin design of experiments (DOE) samples geometrically admissible layouts. A 70 N quasi-static process load is applied over the microcontroller footprint, and a layout is retained when the maximum equivalent elastic strain in the ceramic resistor strain interest zone stays below 1600 $\mu\varepsilon$. All statically feasible layouts then proceed to modal and harmonic analysis at a critical plated through hole component's (VDR3) lead hotspot, where acceptance requires the largest PCB dominated stress peak below 5% of the component dominated peak and the first PCB dominated mode above 800 Hz. Of 400 sampled layouts, 15 (3.75%) pass the static screen and five meet both dynamic requirements; the smallest sufficient mount count is four. The selected four-mount layout attains $J_L = 1499.2$ $\mu\varepsilon$ and $f_{\mathrm{PCB},1} = 823.6$ Hz with no detected PCB-dominated stress peak. Static screening yields a compact, mount count specific feasible pool that concentrates dynamic analysis on structurally credible layouts.

I. Introduction and Research Gap

Permanent mounts on printed circuit boards (PCBs) are generally constrained by package geometry, electrical keep-out regions, and manufacturing access. Their locations also shape the load path during assembly and the flexural coupling between the board and the component leads. Recent PCB mount-layout optimization has primarily addressed resonance avoidance by selecting mounting locations and mount count to improve a global natural-frequency response [1, 2]. Furthermore, contemporary literature emphasizes the critical role of boundary condition optimization and layout configuration in managing localized thermomechanical stress and dynamic vibration fatigue in complex assemblies [3, 4, 5, 6]. To navigate these increasingly complex design spaces, researchers frequently employ heuristic and machine learning algorithms for spatial layout and support optimization [7, 8]. However, a global natural-frequency objective alone does not determine whether a mount pattern is acceptable under a localized process load, nor whether PCB-dominated modes contribute materially to stress at a critical component lead. This study addresses that gap with a traceable, staged proposal framework. Spatially diverse layouts are sampled from a constrained candidate mount set,

avoiding exhaustive enumeration; a static process strain stage then identifies a feasible pool for each mount count, and a dynamic stage then evaluates every layout in that pool at the heavy Varistor's (VDR3) lead hotspot. The recommended mount count is the smallest for which an evaluated layout satisfies both the lead stress contribution and first PCB mode frequency requirements.

II. Staged Computational Proposal Framework

A. Geometric feasibility and stratified maximin DOE

Let $C = \{c_1, \ldots, c_{N_c}\}$ be the set of N_c allowable candidate mount locations. Each layout is a binary vector $z \in \{0,1\}^{N_c}$, where $z_i = 1$ denotes that candidate c_i is active. Let $N_{\min}$ and $N_{\max}$ bound the number of active mounts, $C_r \subseteq C$ the candidates in allowed region r, d_{ij} the separation between c_i and c_j, and $d_{\min}$ the minimum permitted inter-mount spacing. The geometrically admissible set is

$$\mathcal{X}_G = \left\{ z \in \{0,1\}^{N_c} \; : \; \begin{array}{l} N_{\min} \le \sum_{i=1}^{N_c} z_i \le N_{\max}, \\ \sum_{i \in C_r} z_i \le 1 \;\; \forall r, \\ z_i + z_j \le 1 \;\; \forall (i,j) : d_{ij} < d_{\min} \end{array} \right\}. \tag{1}$$

To represent $\mathcal{X}_G$ without evaluating every admissible combination, an information-weighted stratified maximin DOE was used [**?**]. The 400 sampled layouts comprise 140 with two active mounts, 108 with three, 88 with four, 44 with five, and 20 with six. The larger allocation at lower mount counts provides denser coverage where placement is expected to exert greater topological influence on the response. The reported outcome is therefore the best layout within the sampled DOE, not a global optimum over $\mathcal{X}_G$.

B. Static process strain screen

For each sampled layout, a 70 N load is distributed over the microcontroller contact footprint. Let $\mathcal{R}$ denote the set of protection zones, Ω_r the domain of zone $r \in \mathcal{R}$, and $q(\boldsymbol{x}; z)$ the equivalent von-Mises elastic strain at location $\boldsymbol{x}$ for layout z. The static screening quantity is the maximum equivalent strain over all protection zones,

$$J_L(z) = \max_{r \in \mathcal{R}} \max_{\boldsymbol{x} \in \Omega_r} q(\boldsymbol{x}; z). \tag{2}$$

The static feasible set is

$$\mathcal{X}_S = \{ z \in \mathcal{X}_{\mathrm{DOE}} : J_L(z) \le 1600 \, \mu\varepsilon \}. \tag{3}$$

The criterion is deliberately conservative, using the worst local equivalent strain rather than a zone average or global response, so that it rejects mount patterns producing an unacceptable local load path even when the overall board response appears modest.

C. Mode resolved dynamic screen

The dynamic assessment set is the full static-feasible set, $\mathcal{X}_D = \mathcal{X}_S$. For every layout in $\mathcal{X}_D$, modal strain energy is partitioned among the PCB, the component body, and the lead. A mode is PCB-dominated when at least 60% of its strain energy resides in the board while the component-body and lead shares stay below 40% and 25%, respectively; it is component-dominated when the component-body share is at least 40% or the lead share is at least 25%; the remaining modes are coupled. The first PCB-dominated natural frequency is $f_{\mathrm{PCB},1}$.

A $1g$ harmonic base acceleration sweep from 10 to 2000 Hz with a 1% damping ratio, is applied at each active mount along the heavy component's critical failure axis. Each layout is assessed based on the peak RMS von Mises stress at the VDR3's lead hotspot. Peaks in this stress spectrum are assigned to the nearest classified modes. With $\hat{\sigma}_{\mathrm{PCB}}$ and $\hat{\sigma}_{\mathrm{comp}}$ the largest peaks associated with PCB dominated and component dominated modes, the PCB stress contribution is

$$\rho_{\mathrm{PCB}}(z) = 100\,\frac{\hat{\sigma}_{\mathrm{PCB}}(z)}{\hat{\sigma}_{\mathrm{comp}}(z)}\ \%. \tag{4}$$

If no PCB dominated peak is detected while a component dominated peak is present, the layout is plotted at $\rho_{\mathrm{PCB}} = 0$ for this peak-based criterion. The feasible set is

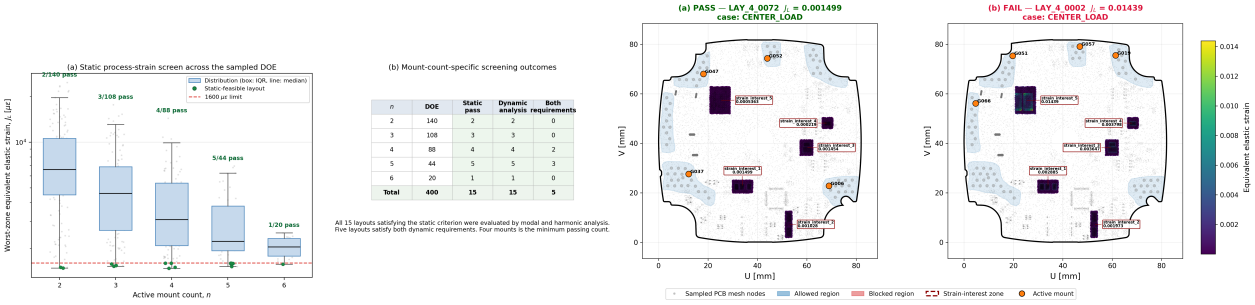
$$\mathcal{X}_F = \left\{ z \in \mathcal{X}_D : \rho_{\mathrm{PCB}}(z) \le 5\%,\ \ f_{\mathrm{PCB},1} > 800\ \mathrm{Hz} \right\}. \tag{5}$$

The selected mount count is the smallest count for which an evaluated layout satisfies every prescribed requirement, $n^* = \min_{z \in \mathcal{X}_F} \sum_{i=1}^{N_c} z_i$.

III. Numerical Results

Figure **??** separates the static process strain screen from the subsequent dynamic assessment. Under $J_L \le 1600\ \mu\varepsilon$, 15 of the 400 sampled layouts pass, i.e. 3.75% of the DOE. Retention rates for two through six mounts are 1.4% (2/140), 2.8% (3/108), 4.5% (4/88), 11.4% (5/44), and 5.0% (1/20), respectively. The static stage therefore does more than rank layouts; it isolates a compact, mount count specific pool of spatially

credible patterns. All 15 statically feasible layouts are then carried into the dynamic stage and assessed by modal and harmonic analysis.



n	DOE	Static pass	Dynamic analysis	Both requirements
2	140	2	2	0
3	108	3	3	0
4	88	4	4	2
5	44	5	5	3
6	20	1	1	0
Total	**400**	**15**	**15**	**5**

Figure 1: Static process-strain stage. *Left:* (a) distribution of J_L over all 400 sampled layouts, with green markers satisfying $J_L \leq 1600\ \mu\varepsilon$, and (b) count-wise outcomes of the static screen; all 15 feasible layouts proceed to dynamic analysis. *Right:* equal-count comparison of two four-mount layouts under the same 70 N load—a compliant layout at $J_L = 1499.2\ \mu\varepsilon$ and a rejected layout at $J_L = 14387.5\ \mu\varepsilon$.

Mount placement remains decisive at a fixed support count. The right panel of Fig. 1 compares two four-mount layouts under the same 70 N process load. The selected layout, `LAY_4_0072`, gives $J_L = 1499.2\ \mu\varepsilon$, whereas the rejected equal-count layout, `LAY_4_0002`, gives $J_L = 14387.5\ \mu\varepsilon$an approximately 9.6 fold difference at identical mount count. Support number alone is therefore not an adequate design variable; the static screen identifies feasible *positions* within each stratum, not merely a sufficient count.

The final count cannot be chosen from static strain alone. Combining the count-wise results of Fig. 1 with the acceptance map of Fig. 2: at two mounts, both layouts fail both requirements ($\rho_{\mathrm{PCB}} > 5\%$ and $f_{\mathrm{PCB},1} < 800$ Hz). At three mounts, no PCB dominated harmonic peak is detected, yet the highest first PCB-dominated frequency is only 514.3 Hz. At four mounts, two layouts—`LAY_4_0072` and `LAY_4_0004`—satisfy both requirements. Three of the five five mount layouts also pass but do not reduce the minimum sufficient count, whereas the single six mount layout stays below threshold at $f_{\mathrm{PCB},1} = 783.4$ Hz. Additional supports are thus neither inherently necessary nor inherently sufficient: across the 15 dynamically assessed layouts, four mounts is the minimum sufficient count, $n^* = 4$.

Among the two feasible four-mount layouts, `LAY_4_0072` (mounts G006, G037, G047, G052) is selected because, it satisfy the static strain criterion and first PCB mode frequency margins. Its static margin is 100.8 $\mu\varepsilon$ (6.30%) and its frequency margin is 23.6 Hz (2.95%); no PCB dominated harmonic stress peak is detected. The nearest sampled frequency to the first PCB-dominated mode is 825.9 Hz, where the RMS von Mises stress is only 0.10% of the component-dominated peak. The right panel of Fig. 2 shows the component-dominated peak at the ≈ 229.5 Hz lead-dominated mode, while the first PCB-dominated mode occurs at 823.6 Hz and

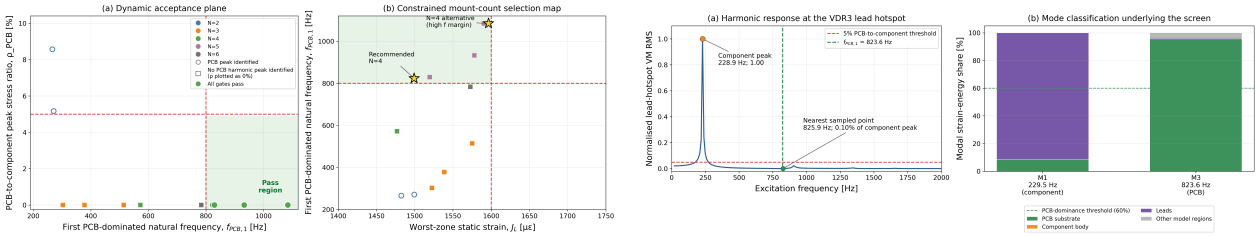


Figure 2: Dynamic stage for the static-feasible layouts. *Left:* (a) the acceptance plane applying $\rho_{\mathrm{PCB}} \leq 5\%$ and $f_{\mathrm{PCB},1} > 800$ Hz, where square markers denote layouts with no detected PCB-dominated peak (plotted at $\rho_{\mathrm{PCB}} = 0$), and (b) the same layouts against the static-strain and first-PCB-mode frequency requirements, with the green region marking simultaneous satisfaction and filled markers fully compliant layouts. *Right:* mode-resolved evidence for `LAY_4_0072`—the component-dominated lead-hotspot peak at ≈ 229.5 Hz with no PCB-dominated peak, and the modal strain-energy partitions used for classification.

carries 95.5% of its strain energy in the board.

The alternative `LAY_4_0004` offers a much larger frequency margin ($f_{\mathrm{PCB},1} = 1084.4$ Hz) but only 3.5 $\mu\varepsilon$ of static headroom, and is retained as a frequency-robust four-mount option. The closest lower-count comparator, `LAY_3_0096`, is statically compliant ($J_L = 1575.2$ $\mu\varepsilon$) with no detected PCB-dominated peak, but fails the frequency requirement ($f_{\mathrm{PCB},1} = 514.3$ Hz). The five-mount comparator `LAY_5_0036` also satisfies the requirements but does not improve the minimum-count decision. Because the selected layout clears the frequency threshold by only 23.6 Hz, it is the primary candidate for a manufacturing-tolerance study.

IV. Conclusions and Future Work

The 1600 $\mu\varepsilon$ static screen reduces the 400-layout DOE to a 15-layout feasible pool, and all 15 layouts in this pool undergo modal and harmonic analysis. Four mounts is the smallest count that simultaneously satisfies the 5% PCB-contribution limit at the lead hotspot and $f_{\mathrm{PCB},1} > 800$ Hz; the recommended layout, `LAY_4_0072`, attains $J_L = 1499.2$ $\mu\varepsilon$ and $f_{\mathrm{PCB},1} = 823.6$ Hz with no detected PCB-dominated stress peak. These conclusions apply to the sampled DOE, rather than constituting a global optimum over all admissible mount layouts.

Future work will train a static-response surrogate model using the 400 completed FE simulations to predict the worst-zone equivalent elastic strain, J_L, across a broader design space. By identifying static-feasible layouts, this approach reserves computationally expensive modal and harmonic analyses for structurally credible candidates, building on established surrogate-assisted optimization methods for mixed-variable problems [9]. Finally, a manufacturing-tolerance study will quantify the robustness of the selected layout, specifically its 23.6 Hz frequency margin.

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